PYTHON FOR DATA ANALYST | CASE STUDY

BukaToko: User Behaviour & Funnel Analysis

From 10,000 user events to a clear growth agenda



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Dataset: 10,000 events, 27 Mar – 23 Sep 2025 | Python (pandas, matplotlib), Google Sheets

EXECUTIVE SUMMARY

Growth is stalled at ~1,150 MAU: the lever is conversion, not acquisition

1,055

active users in Indonesia, the #1 market in Q2 2025 (25.2% of activity)

~1,150

monthly active users, flat for five months (1,129–1,170, Apr–Aug)

+19.7%

referral MAU growth, the only growing channel; social falls −13.0%

89.6%

of sessions on mobile; Android alone is 49.6%

53.2%

of product viewers add to cart: the biggest leak in the funnel

0.13%

same-day search-to-cart, but 38.1% of searchers add to cart on another day

CONTEXT

The case: a new analyst on the BukaToko analytics team

A regional e-commerce platform

Web and mobile app, operating across Southeast Asia, the US and Australia.

A familiar shopper journey

Users browse, search, view products, add to cart and check out; some buy, some drop off.

Six acquisition channels

Organic search, paid ads, email, social media, referrals and direct visits.

My job

Explore user behaviour in the event-level data and find where the funnel can improve.

QUESTIONS TO ANSWER

Q1 Which country has the most active users in Q2 2025?

Q2 How is the MAU trend for each channel?

Q3 Which device is used most often, by session?

Q4 What % of users who search add to cart the same day?

DATA

10,000 events from 1,982 users, cleaned and verified before analysis

10,000
events (1 event per session)

1,982
distinct users

8
countries and 8 event types

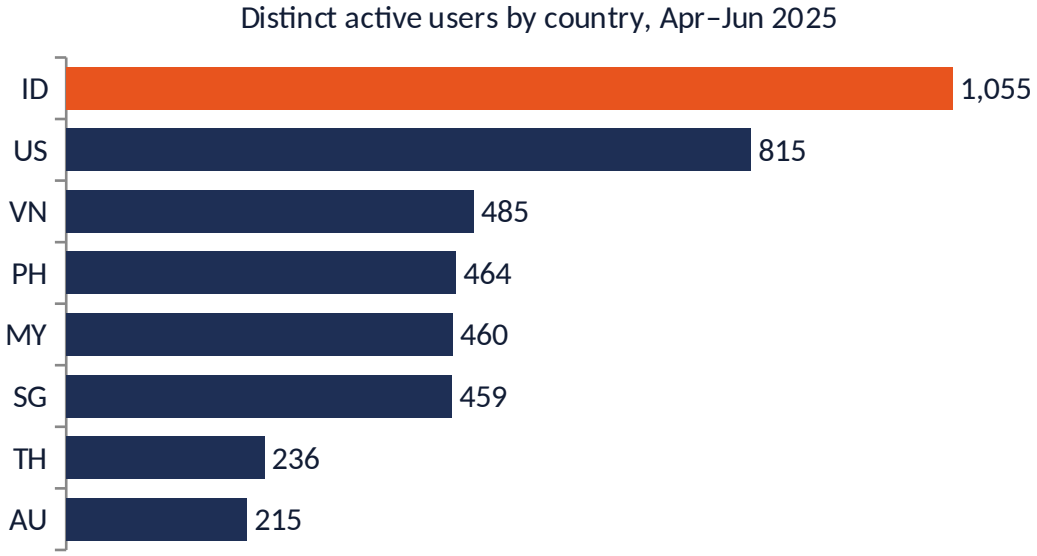
6 months
27 Mar – 23 Sep 2025

Cleaning applied

- Device casing normalised (“IOS”, “android”)
- 100 blank channels mapped to “unknown”
- Non-padded hours parsed; all timestamps valid
- No duplicate rows or event IDs; no missing IDs
- product_id empty on 5,537 non-product events (expected)

QUESTION 1

Indonesia has the most active users in Q2 2025



1,055

active users in Indonesia, out of 1,848 distinct users active in Q2

44.7%

of country activity comes from ID + US

Answer: Indonesia (ID), with 1,055 distinct active users and 1,490 events in Apr–Jun 2025, ahead of the US (815).

Two market tiers, and Southeast Asia dominates the rest

Country	Active users	Events	Share
ID	1,055	1,490	25.2%
US	815	1,032	19.5%
VN	485	541	11.6%
PH	464	526	11.1%
MY	460	515	11.0%
SG	459	521	11.0%
TH	236	249	5.6%
AU	215	231	5.1%

Method note: country is stored per event and most users appear under more than one country, so a user is counted once in each. Shares are of 4,189 user-country pairs.

Two clear tiers

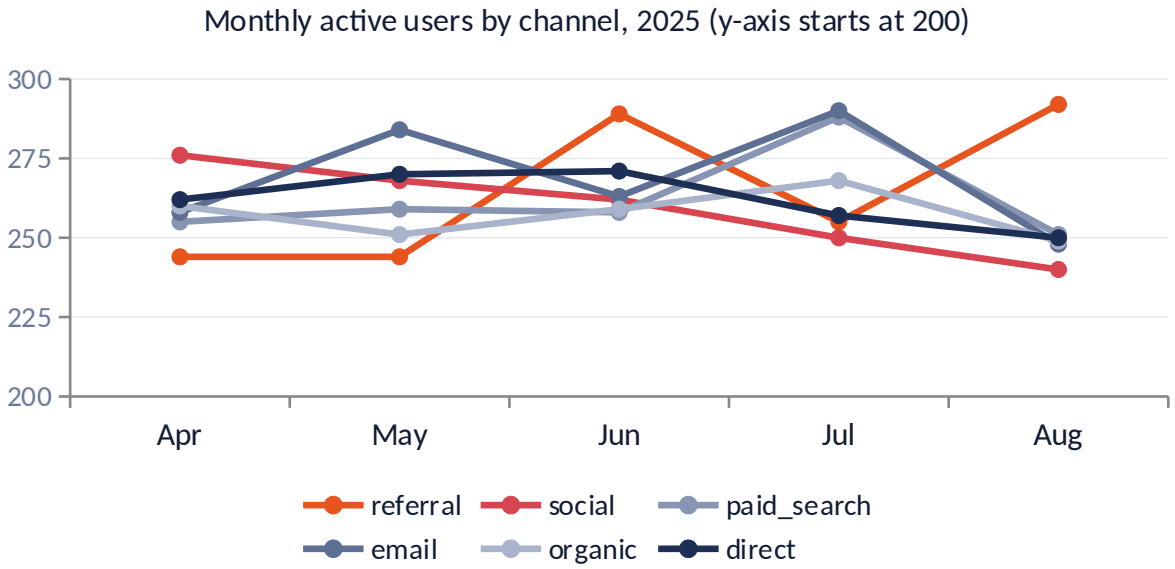
ID and US lead. VN, PH, MY and SG form a tight middle band (459–485). TH and AU trail.

SEA is the core

Six of eight markets are in Southeast Asia, consistent with BukaToko's home region.

QUESTION 2

MAU per channel: flat overall, referral grows, social declines



1,129–1,170

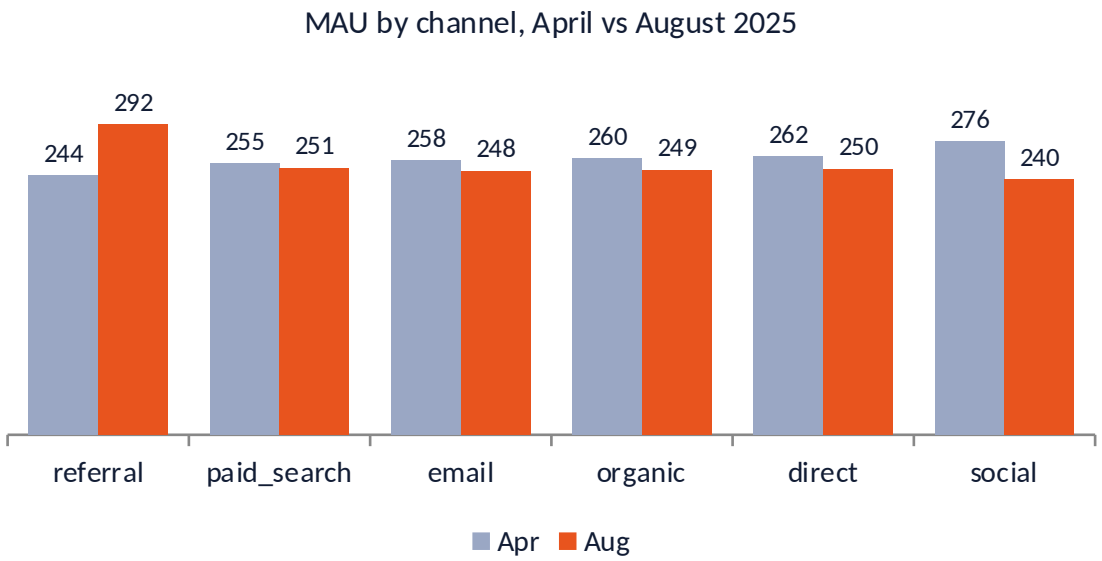
total MAU per month, Apr–Aug

No channel breaks away: each holds roughly 240–290 MAU.

Answer: every channel is essentially flat across Apr–Aug. Referral is the only grower (+19.7%); social falls the most (−13.0%).

QUESTION 2 | DETAIL

Referral overtakes social as the #1 channel by August



Referral +19.7%

Up from 244 to 292 (uneven month to month), now the #1 channel.

Social −13.0%

From the top channel in April (276) to the bottom of the main group (240).

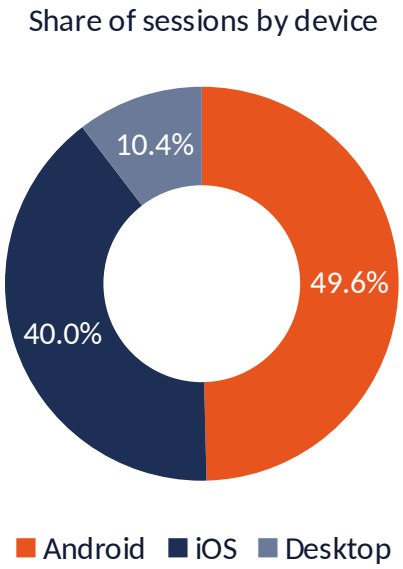
Others −1.6% to −4.6%

Paid search, email, organic and direct are flat: acquisition is not compounding.

Mar and Sep are partial months in the data (from 27 Mar, to 23 Sep), so their lower MAU is a coverage effect. "unknown" is omitted (tiny base: 15 → 20).

EXTRA QUESTION 3

Android is the most used device, and mobile is nearly 90% of sessions



89.6%

of sessions are on mobile (Android 49.6% + iOS 40.0%)

Sessions: Android 4,961 | iOS 4,000 | Desktop 1,039

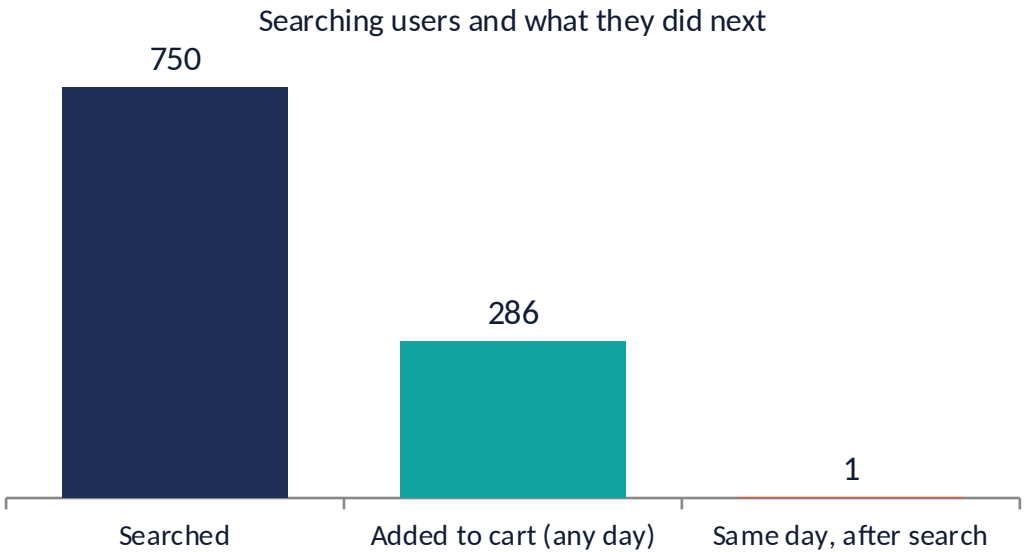
Android leads everywhere It is the top device in all 8 markets (47–52% of events per country).

Desktop is niche 10.4% overall and only 9–12% in any single country.

Answer: Android, 4,961 of 10,000 sessions (49.6%). Each session holds one event, so session share equals event share.

EXTRA QUESTION 4

Almost no one searches and adds to cart on the same day: 0.13%



0.13%

1 of 750 searching users added to cart the same day

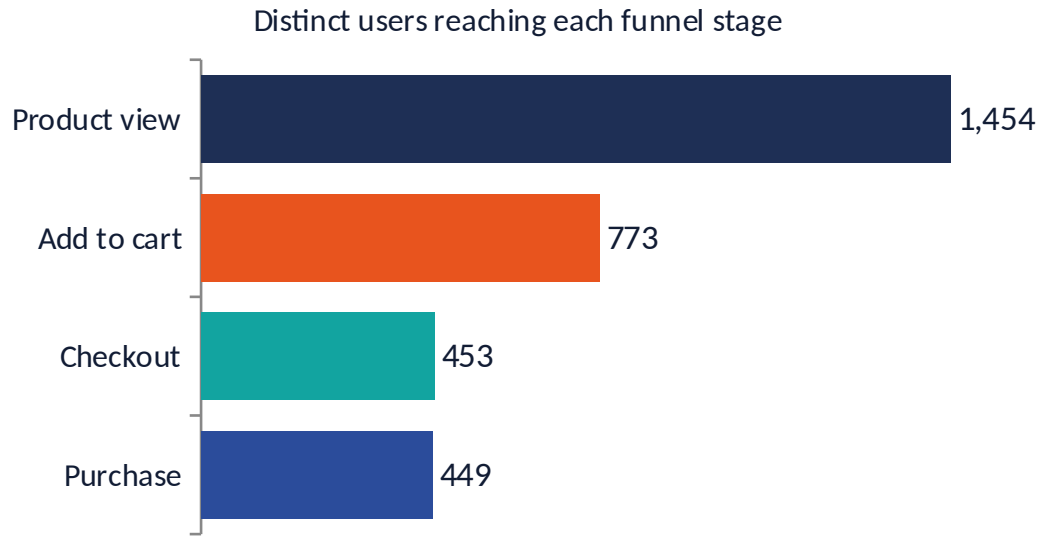
The rule

Add to cart must come after a search, same user, same date. Only 2 user-days contain both events.

Why so low? users average 5.05 events in ~6 months, so a search and a cart add rarely share a day. 38.1% add to cart on another day.

FUNNEL ANALYSIS

The biggest leak: nearly half of product viewers never add to cart



53.2% — **Product view → Add to cart**
−46.8% drop: the biggest leak

58.6% — **Add to cart → Checkout**
−41.4% drop: second leak

99.1% — **Checkout → Purchase**
Checkout is almost frictionless

Overall: 449 of 1,454 product viewers (30.9%) purchase. Fix view → cart first, then cart → checkout.

User-level counts: distinct users with at least one event of that type. Stages are not strictly sequential per user.

RECOMMENDATIONS

Six moves to turn flat MAU into growth

HIGH

Fix the cart gap

Test product-page changes to lift view → cart above 53.2%.

HIGH

Back Indonesia

Prioritise ID assortment, payments and logistics (25.2% of activity).

HIGH

Retain, don’t just acquire

MAU is flat at ~1,150: invest in conversion and re-engagement.

MEDIUM

Scale referral

The only growing channel (+19.7%): build a referral programme.

MEDIUM

Fix or shift social

Down −13.0%: review creative and targeting, or shift budget.

MEDIUM

Mobile-first, Android-first

89.6% of sessions are mobile; Android alone is half.

NEXT STEPS | PROPOSED

A 90-day roadmap: measure, test, then scale

30 Measure
days 1–30

- Baseline KPIs: MAU, view → cart, cart → checkout
- Adopt a 7/30-day search-to-cart window
- Audit social creative and targeting

60 Test
days 31–60

- A/B test product-page changes for view → cart
- Pilot a referral programme in Indonesia
- Android-first review of key flows

90 Scale
days 61–90

- Roll out winning tests to US and SEA markets
- Rebalance channel budget on attribution data
- Track retention cohorts monthly

KPIs to watch MAU | view → cart rate | cart → checkout rate | referral share of MAU | 7/30-day search → cart rate

METHODOLOGY

Definitions, rules and limitations

Active user	Distinct user_id with at least one event in the period. Q2 2025 = 1 Apr to 30 Jun.
MAU	Distinct users with at least one event in the month carrying that channel; multi-channel users count in each.
Session (Q3)	Distinct session_id per device. In this dataset each session holds exactly one event.
Same-day conversion (Q4)	Add to cart strictly after the user's first search that date. 1 of 945 search days = 0.11% per user-day.

Limitations

- Country and channel are per-event, so totals overlap across users
- March and September are partial months
- Funnel stages are not strictly sequential per user
- Figures cross-checked with an independent SQL recount: exact match

APPENDIX

Code and supporting files

Python code

bukatoko_analysis.py (pandas + matplotlib) reproduces every number and chart from the raw CSV: cleaning rules, groupby aggregations for Q1–Q3 and a first-search vs add-to-cart join for Q4.

Calculations

The supporting Google Sheet holds the Q1 country table, Q2 MAU matrix, Q3 device table and Q4 attribution summary, all from the same cleaned data.

Verification

Headline figures were recomputed independently in SQL (sqlite) from the cleaned dataset; all matched the pandas results exactly.

Files bukatoko_analysis.py | clean_events.csv | dirty_dummy_events.csv | charts/

THANK YOU

Questions & discussion



Indonesia is the engine, referral is the growth signal, and the cart gap is the fix worth making first.

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